

Vridhi Brahmhatt

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SUMMARY

Full-Stack Engineer specializing in Python (Django) and TypeScript (Next.js) with a focus on complex business logic, cryptographic auditing, and financial systems. Proven ability to translate rigorous requirements into reliable, well-tested backends and intuitive frontends, as demonstrated by independent architecture of a RWA tokenization protocol and a security monitoring platform. Active open-source contributor focused on FinTech and Web3 infrastructure.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, Solidity, SQL, Bash, HTML/CSS
Backend:	Django, DRF, PostgreSQL, Redis, REST API Design, GraphQL, JWT Auth
Frontend:	Next.js, React, TailwindCSS, Vite, Wagmi, TanStack Query
Blockchain:	Foundry, Ethereum, LayerZero, Chainlink, ERC-4337, Smart Contract Dev
Testing & QA:	Unit, Integration, Fuzz, & Invariant Testing, Risk Scoring Algorithms
DevOps & Infra:	Git, GitHub Actions, Linux (LPI Essentials), Systemd Automation, Rate Limiting, CI/CD Pipelines

KEY PROJECTS

Cryptographic Point-of-Capture Ledger (Seal) | *Next.js, TypeScript, Django, Python, TailwindCSS*

- Architected a tamper-proof evidence logging system utilizing a cryptographic hash chain (SHA-256), making retroactive data alteration mathematically impossible without requiring a traditional blockchain.
- Engineered a strict state-machine workflow (Handoff → Disputed → Resolved) with an ORM-level immutability lock to ensure regulatory compliance during disputes.
- Developed an Optimistic UI with a delayed-commit undo pattern (5s countdown) to eliminate accidental state changes in high-stress physical environments.
- Designed dual-context interfaces: a high-contrast Kiosk for physical tablet capture, and a dense, dark-mode analytics Dashboard.
- Implemented a zero-trust client-side audit feature using the native Web Crypto API to independently verify data integrity directly in the browser.

LayerZero Infrastructure Security Scanner | *Django, Next.js, PostgreSQL, Redis, Discord API*

- Built a full-stack monitoring platform that scans external infrastructure configurations for security risks, processes them with a custom risk-scoring engine, and alerts users via webhooks.
- Designed RESTful API with Django REST Framework, JWT auth, rate limiting (Redis), and public/private endpoints.
- Created a responsive Next.js dashboard with pagination, filtering, and search; handled complex state management and API integration with Axios.
- Implemented automated security reports and a user-facing alert channel system to translate complex risk vectors into actionable compliance scores (ranging from 0-100).

Aurum Protocol: Tokenized Gold System | *Solidity, Foundry, Next.js, Chainlink*

- Architected a full-stack application simulating Real World Asset (RWA) tokenization, where users deposit physical assets to mint on-chain tokens, featuring a public dashboard displaying live TVL and risk metrics.
- Designed a dynamic risk model (LTV, interest rates, automatic debt allocation, liquidation curves) based on real-time data feeds, demonstrating complex business logic and financial calculations.
- Wrote extensive automated tests (unit, fuzz, invariant) achieving 95% coverage; performed a self-audit and fixed multiple critical business-logic edge cases.
- Built an interactive Next.js frontend with responsive design; handled wallet connections, transactions, and contract state.

EXPERIENCE

Open Source Contributor

OpenBB, exchange_calendars, OpenZeppelin

April 2026–Present

- **OpenBB:** Identified and filed a detailed bug report for a silent failure in the V5 codegen workflow parsing modular OpenAPI 3.1.0 specs, leading to a fast resolution by the core team.

- **exchange_calendars:** Investigated Shanghai Stock Exchange (XSHG) holiday schedules and identified a separate bug in the CSV generation script related to a deprecated pandas 2.2+ argument.
- **OpenZeppelin:** Submitted documentation PRs to improve developer onboarding and security, including adding a Denial of Service (DoS) warning to TimelockController.

CERTIFICATIONS

- ITIL Foundation Certificate in IT Service Management
- Linux Professional Institute (LPI) Linux Essentials

EDUCATION

Bachelor of Science in Computer Science, Western Governor's University

September 2025